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NETWORK INFORMATION AND SECURITY

Wireshark

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Introduction to Wireshark

The World's Most Popular Network Protocol Analyzer

Open-source — Free to use, maintained by a global community since 1998

Protocol Support — Understands 3,000+ network protocols

Cross-platform — Works on Windows, macOS, and Linux

Live + Offline — Capture packets live or analyze saved .pcap files

What is a Data Packet?

A packet is a small unit of data transmitted over a network.
Large files are broken into many packets, sent independently, and reassembled at the destination.

Packet Anatomy:

Header	Payload	Trailer
Source & Destination IP, protocol info	Actual data being transmitted	Error-checking (checksum)

Key Facts

- Packets are typically 64–1500 bytes (MTU)
- Each packet carries sequence numbers for reassembly
- TCP/IP is the dominant packet protocol on the internet
- Different packets may take different routes to the same destination

Why Use Wireshark?

Troubleshoot Network Issues

Identify dropped packets, latency spikes, and misconfigured devices quickly.

Security Analysis

Detect suspicious traffic, man-in-the-middle attacks, or data exfiltration attempts.

Protocol Learning

Visualize exactly how TCP, HTTP, DNS, and other protocols behave in practice.

Performance Monitoring

Measure bandwidth usage, response times, and identify bottlenecks.

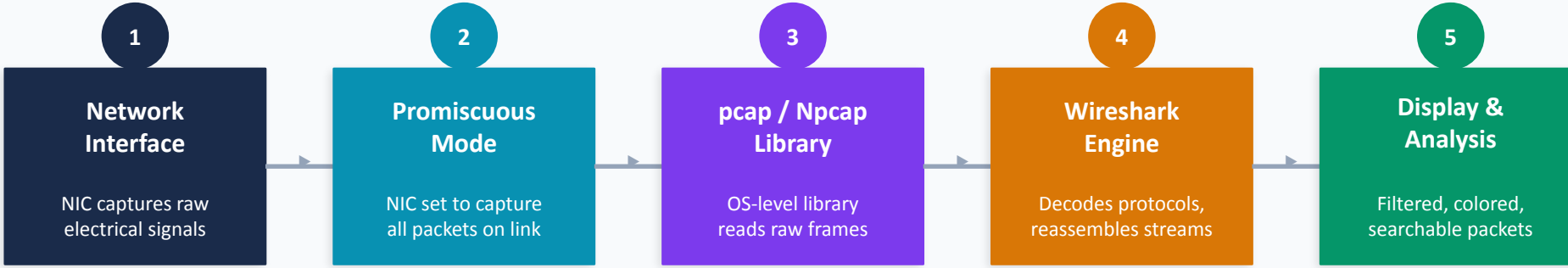
Forensics

Reconstruct sessions and analyze logs after a security incident.

Dev & QA Testing

Verify that your application communicates over the network as expected.

How Wireshark Works



Key Concepts:

Promiscuous Mode — NIC captures all packets on the network segment, not just those addressed to it. **pcap/Npcap** — Low-level library that interfaces between the OS and Wireshark.

Wireshark Interface Overview

① Menu & Toolbar

File Edit View Go Capture Analyze Statistics Help

[Filter: ip.addr==192.168.1.1]

② Filter Bar

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.5	8.8.8.8	DNS	74	Standard query 0x1a2b A google.com
2	0.012451	8.8.8.8	192.168.1.5	DNS	90	Standard query response A 142.250.77.14
3	0.013200	192.168.1.5	142.250.77.14	TCP	66	49812 → 443 [SYN] Seq=0 Win=65535
4	0.028700	142.250.77.14	192.168.1.5	TCP	66	443 → 49812 [SYN, ACK] Seq=0 Ack=1

③ Packet List

```
▼ Frame 1: 74 bytes captured
▼ Ethernet II
▼ Internet Protocol Src: 192.168.1.5
▼ User Datagram Protocol
▼ Domain Name System (query)
```

④ Packet Details

```
0000  45 00 00 4a 00 01 40 00
0008  40 11 f7 6b c0 a8 01 05
0010  08 08 08 08 c9 e4 00 35
0018  00 36 5f 3f 1a 2b 01 00
```

⑤ Hex / Raw Data

Capturing Packets

1

Launch Wireshark

Open the application. The welcome screen shows all available network interfaces.

2

Select Interface

Choose the correct interface (e.g., Wi-Fi, Ethernet). Look for the one with live traffic activity.

3

Start Capture

Click the blue shark-fin icon (▶) or press Ctrl+E to begin capturing packets immediately.

4

Interact with Network

Open a browser, ping a server, or use any app — Wireshark captures all related packets.

5

Stop Capture

Click the red Stop button (⏹) or press Ctrl+E again when done.

6

Save / Analyze

Save as .pcap file (File → Save As) or begin analyzing immediately in the packet list.

Filters and Their Importance

Two Types of Filters:

Capture Filters

Applied before capture begins. Reduces data stored to disk. Uses BPF (Berkeley Packet Filter) syntax.

```
host 192.168.1.1
```

```
port 80
```

```
tcp and port 443
```

```
not icmp
```

Display Filters

Applied after capture. Hides irrelevant packets in the view. Uses Wireshark's own filter language.

```
ip.addr == 192.168.1.1
```

```
http.request.method == GET
```

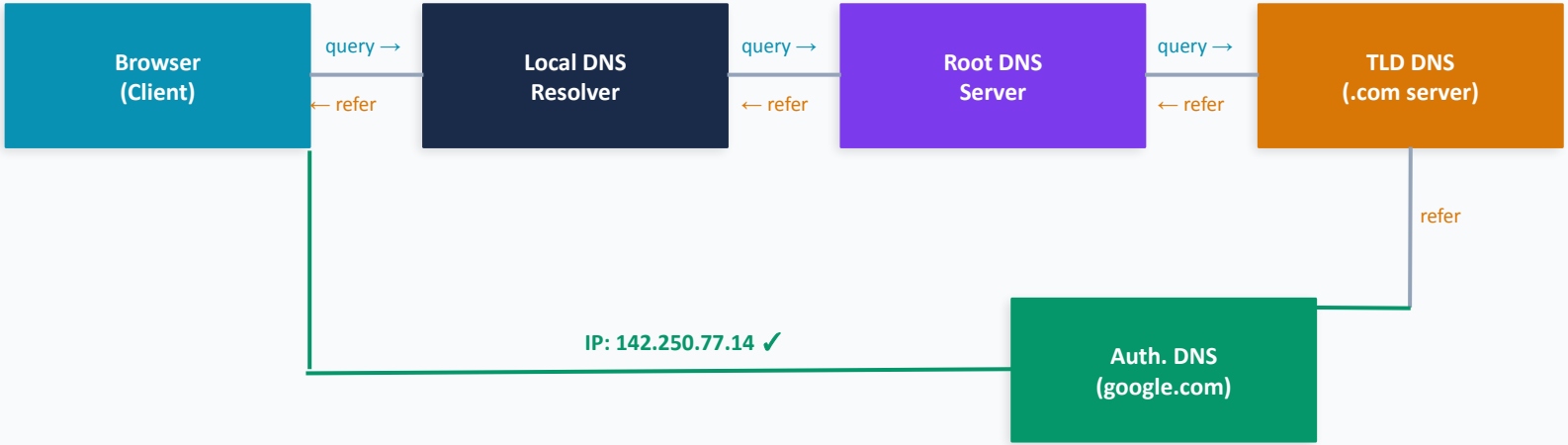
```
dns
```

```
tcp.flags.syn == 1
```

```
frame contains "password"
```

DNS Analysis

DNS Resolution Flow

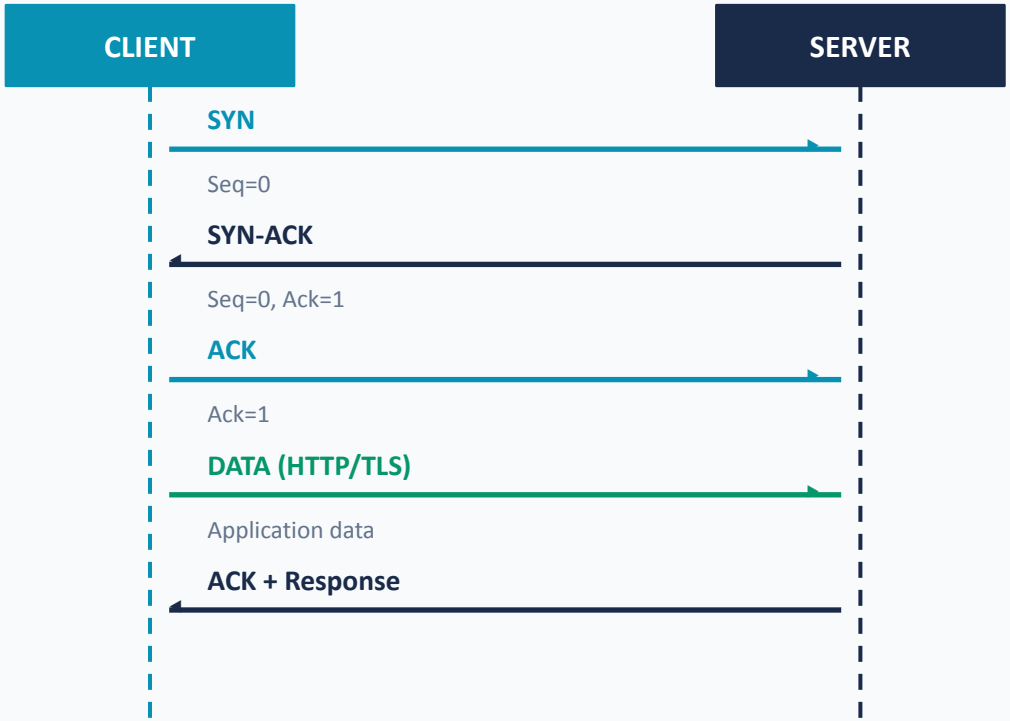


Capturing DNS in Wireshark:

Display filter: `dns` | Follow DNS stream to see query + response | Check `dns.qry.name` for domain lookups | Look for unusually large TXT records (DNS tunneling indicator) | Wireshark captures all these DNS packets in real time. So when this DNS lookup happens in the background, Wireshark records both the DNS query and the DNS response.

TCP Communication Analysis

TCP Three-Way Handshake



TCP in WireShark

- TCP stream reassembly (Follow TCP Stream)
- Detect retransmissions (packet loss)
- Identify duplicate ACKs
- Analyze RTT (latency)
- Check window size (flow control)
- Detect out-of-order packets
- View connection setup & termination
- Use TCP filters (e.g., `tcp.port == 80`)
- Analyze throughput
- Detect connection resets (RST packets)

Demonstration

Real-world Applications

Cybersecurity Incident Response

SOC analysts use Wireshark to investigate breaches — replaying attack traffic, identifying C2 servers, and tracing lateral movement.

Wi-Fi Troubleshooting

ISPs and network engineers capture wireless frames to debug poor connectivity, identify rogues APs, or detect deauthentication attacks.

Application Development

Developers validate that REST APIs, WebSocket connections, or IoT device messages are correctly structured and timed.

Detecting Unencrypted Data

Security auditors identify sensitive data (passwords, tokens) transmitted in plaintext — a critical compliance concern.

VoIP & Video Quality

Telecom engineers analyze RTP/RTCP streams to diagnose jitter, packet loss, and call quality degradation issues.

Malware Analysis

Researchers run malware in sandboxes and capture its network traffic to identify C2 domains, exfiltration channels, or propagation methods.

Conclusion

Wireshark is a powerful, free tool that makes the invisible network visible — one packet at a time.

Understanding packets, protocols, and filters forms the foundation of network analysis and security.

Real-world use spans cybersecurity, performance tuning, protocol learning, and forensics.

Practice with a home lab: capture your own browsing traffic and apply what you've learned today.

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Thank You